```
Parameter Manifest — Hubbard-Holstein L=2

Model:
  - model_family: Hubbard-Holstein
  - problem: hh

Ansatz:
  - ansatz_type: hh_hva
  - vqe_method: SPSA
  - vqe_method_internal: hardcoded_hh_hva_layerwise_statevector
  - initial_state_source: vqe
  - branch_order: exact_gs_filtered, exact_paop, trotter_paop, exact_hva, trotter_hva

Drive Enabled: False
  - spsa_a=0.2  spsa_c=0.1  spsa_A=10.0
  - spsa_alpha=0.602  spsa_gamma=0.101
  - spsa_eval_repeats=1  spsa_eval_agg=mean  spsa_avg_last=0

Branch Provenance:
  - paop_source: internal_adapt
  - paop_pool: uccsd_paop_lf_full
  - paop_depth: 30
  - hva_source: regular_vqe
  - hva_ansatz: hh_hva
  - hva_vqe_success: True
  - adapt_ref_source: vqe

Core Physical Parameters:
  - t=1.0
  - U=4.0
  - dv=0.0

Hubbard-Holstein Parameters:
  - omega0=1.0
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  - g_ep=0.5
  - n_ph_max=1
  - boson_encoding=binary
  - nq_fermion=4
  - nq_phonon=2
  - nq_total=6

Trajectory Settings:
  - t_final=10.0
  - num_times=201
  - trotter_steps=64
  - suzuki_order=2
  - term_order=sorted
  - boundary=periodic
  - ordering=blocked
  - initial_state_source=vqe

Topline Results:
  - subspace_fidelity_at_t0: 0.3625670619623603
  - vqe_energy: 1.7871733127680272
  - exact_filtered_energy: 0.15866790412572704
  - qpe_energy_estimate: None
```
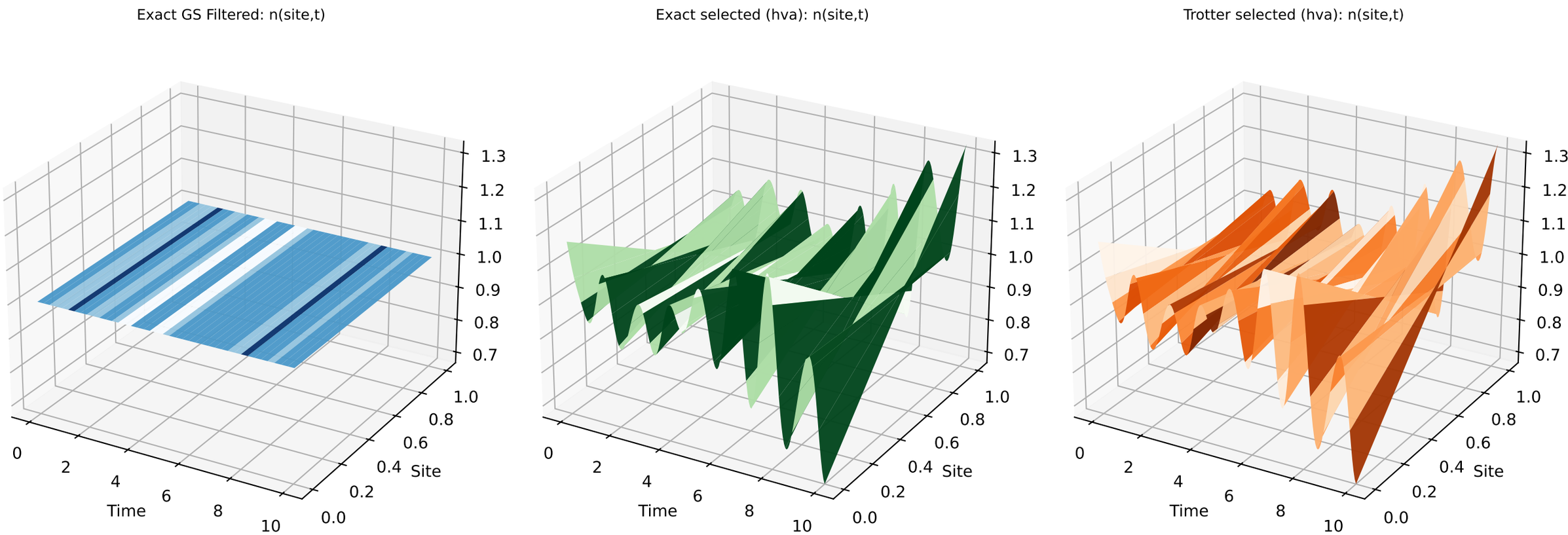
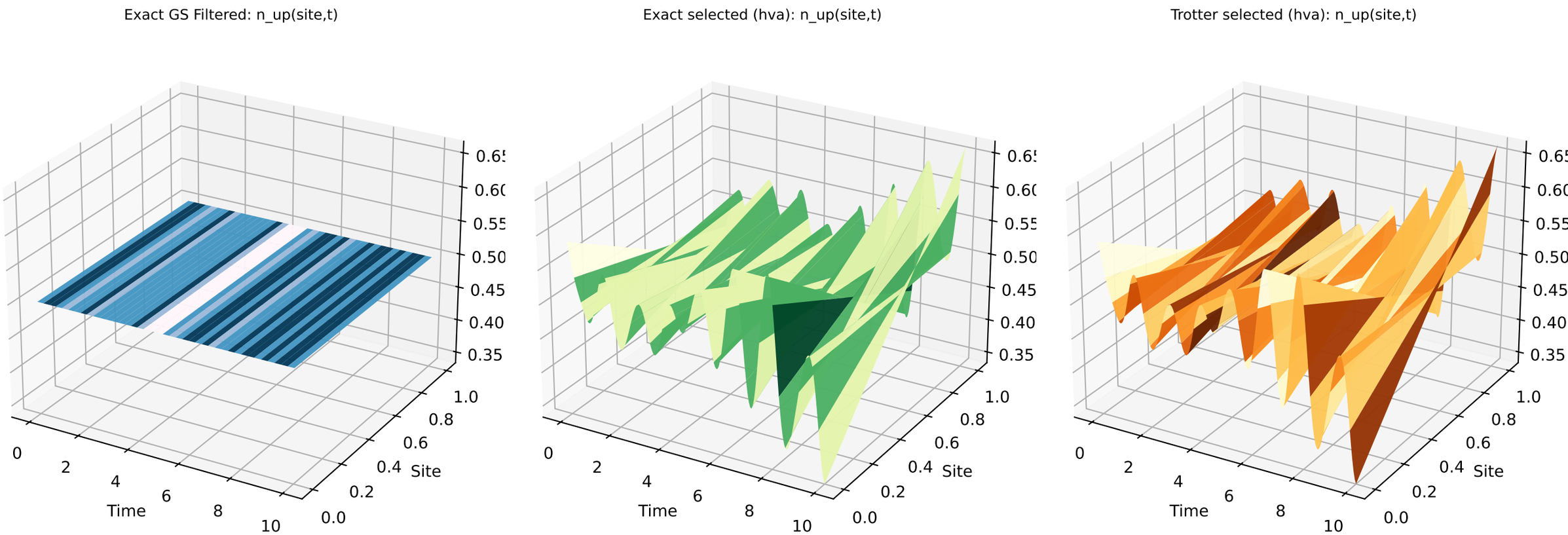
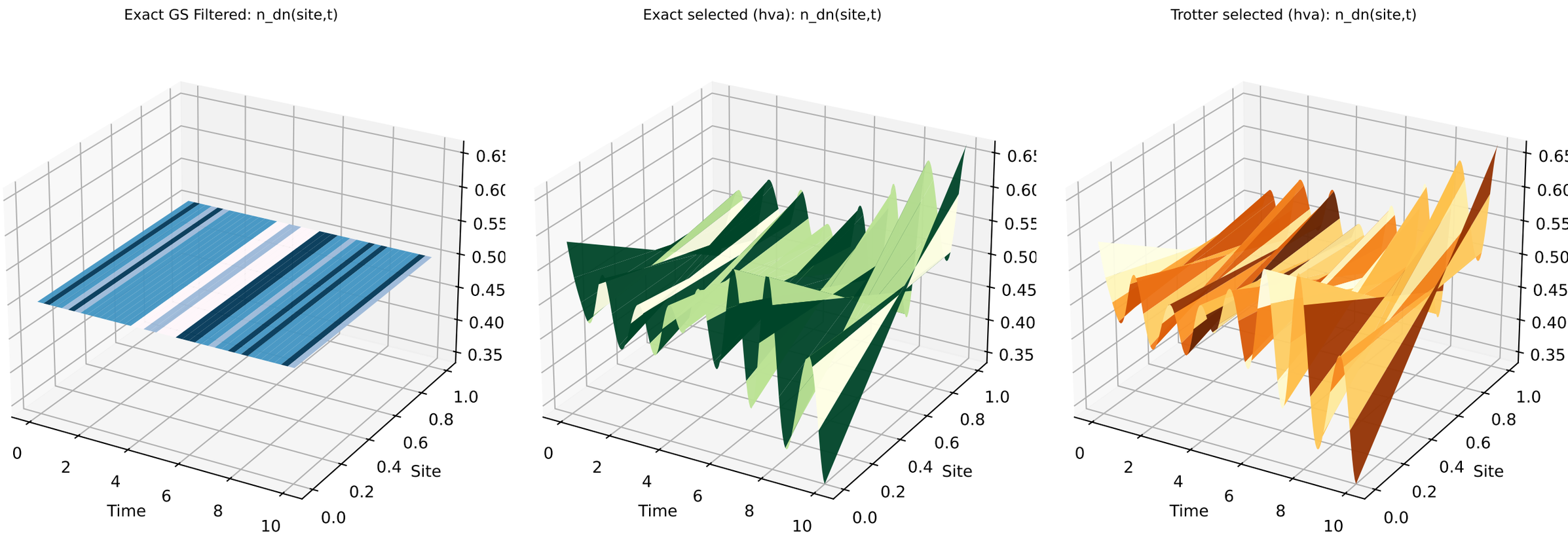

L=2 3D Densities (Total n)



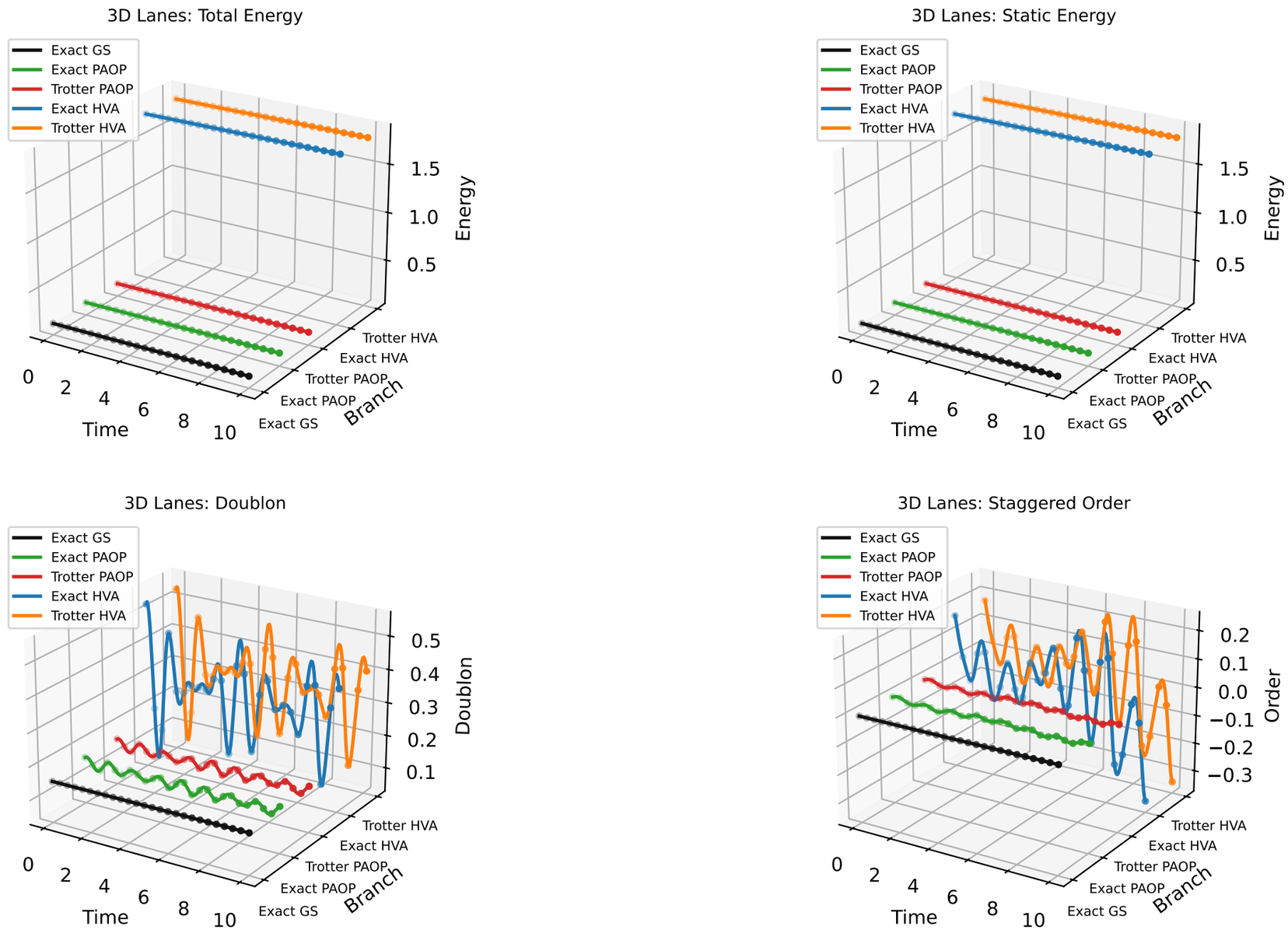
L=2 3D Densities (Spin-Up)

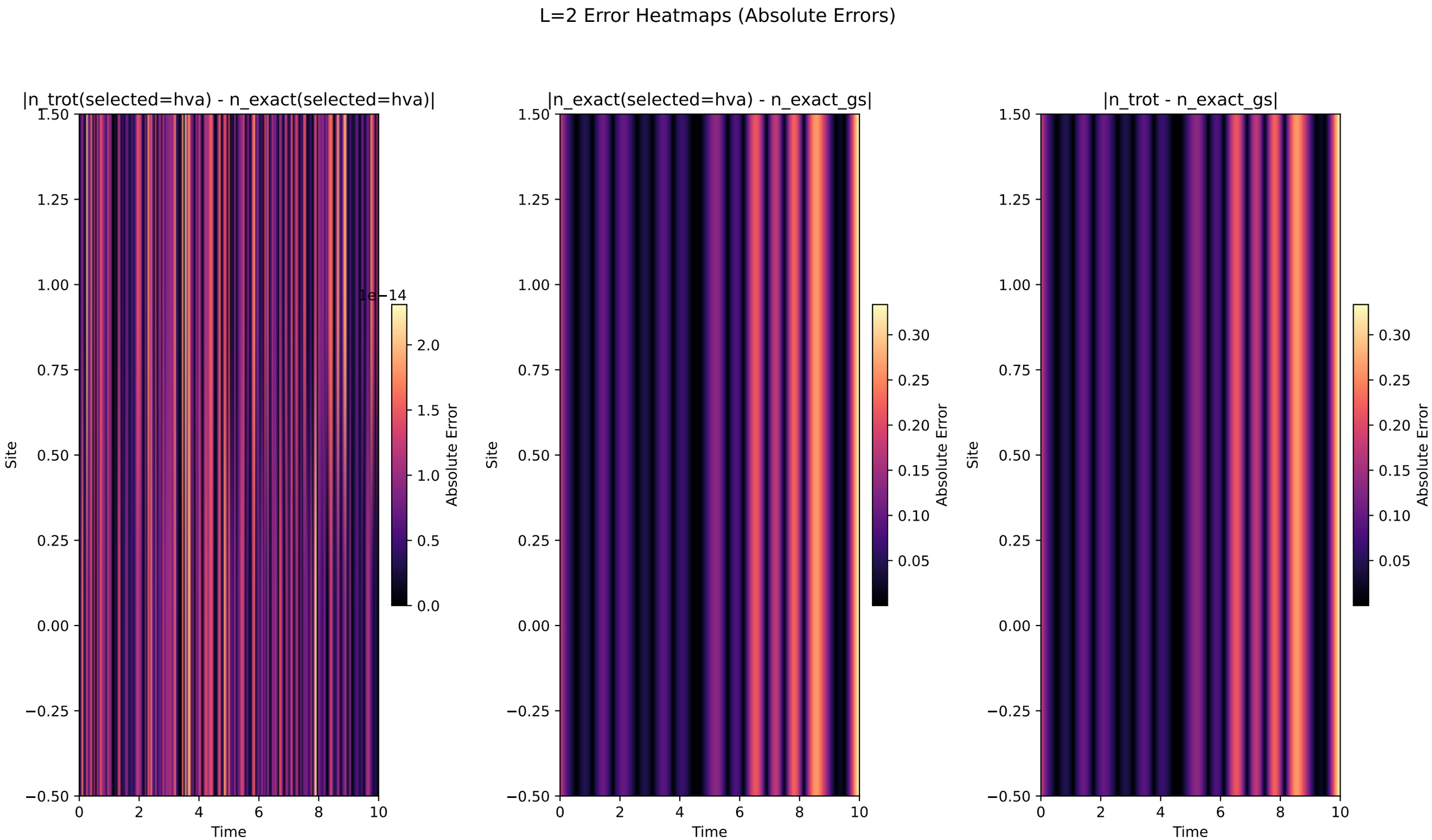


L=2 3D Densities (Spin-Down)

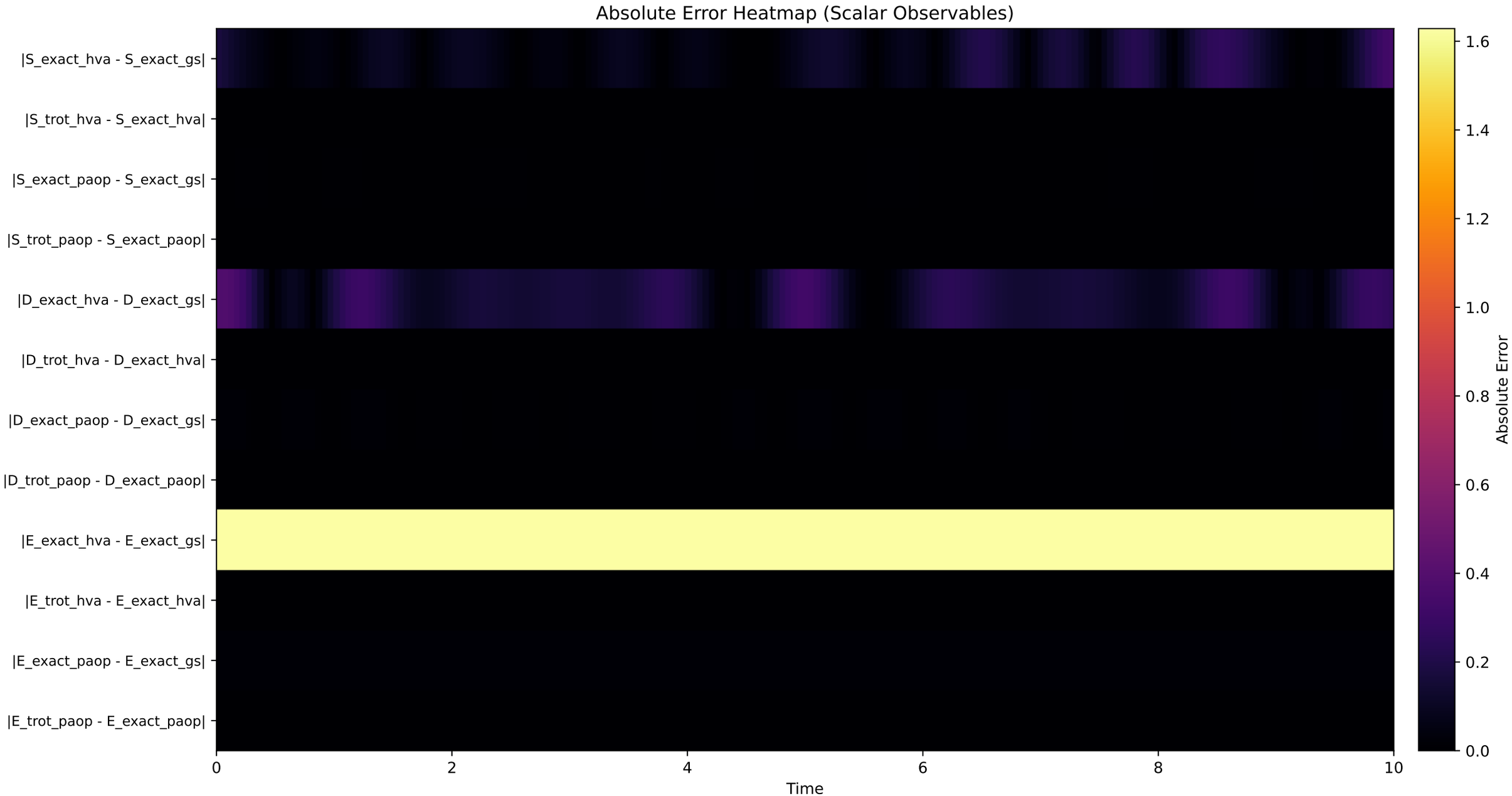


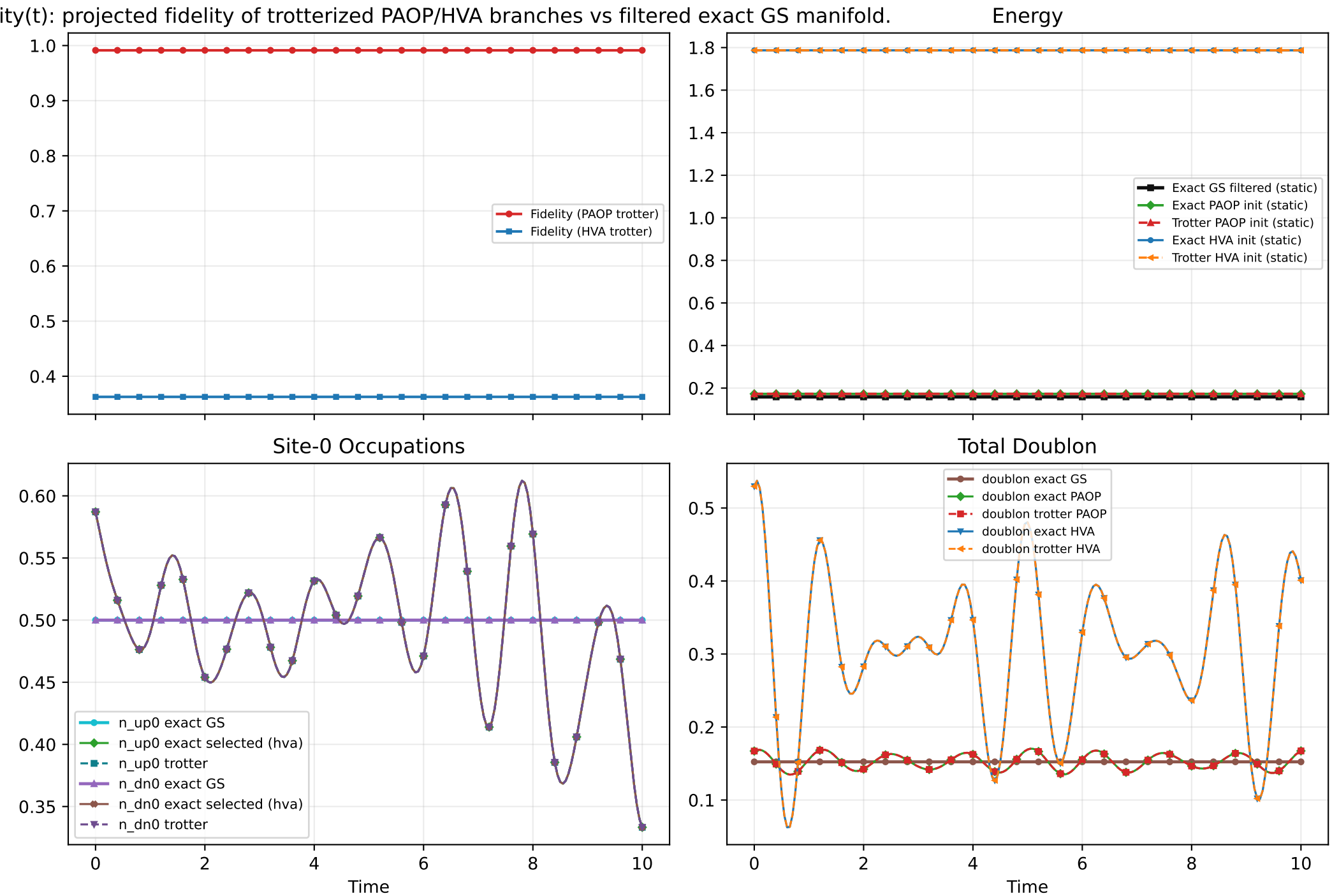
L=2 3D Scalar Observables (Five Branches)

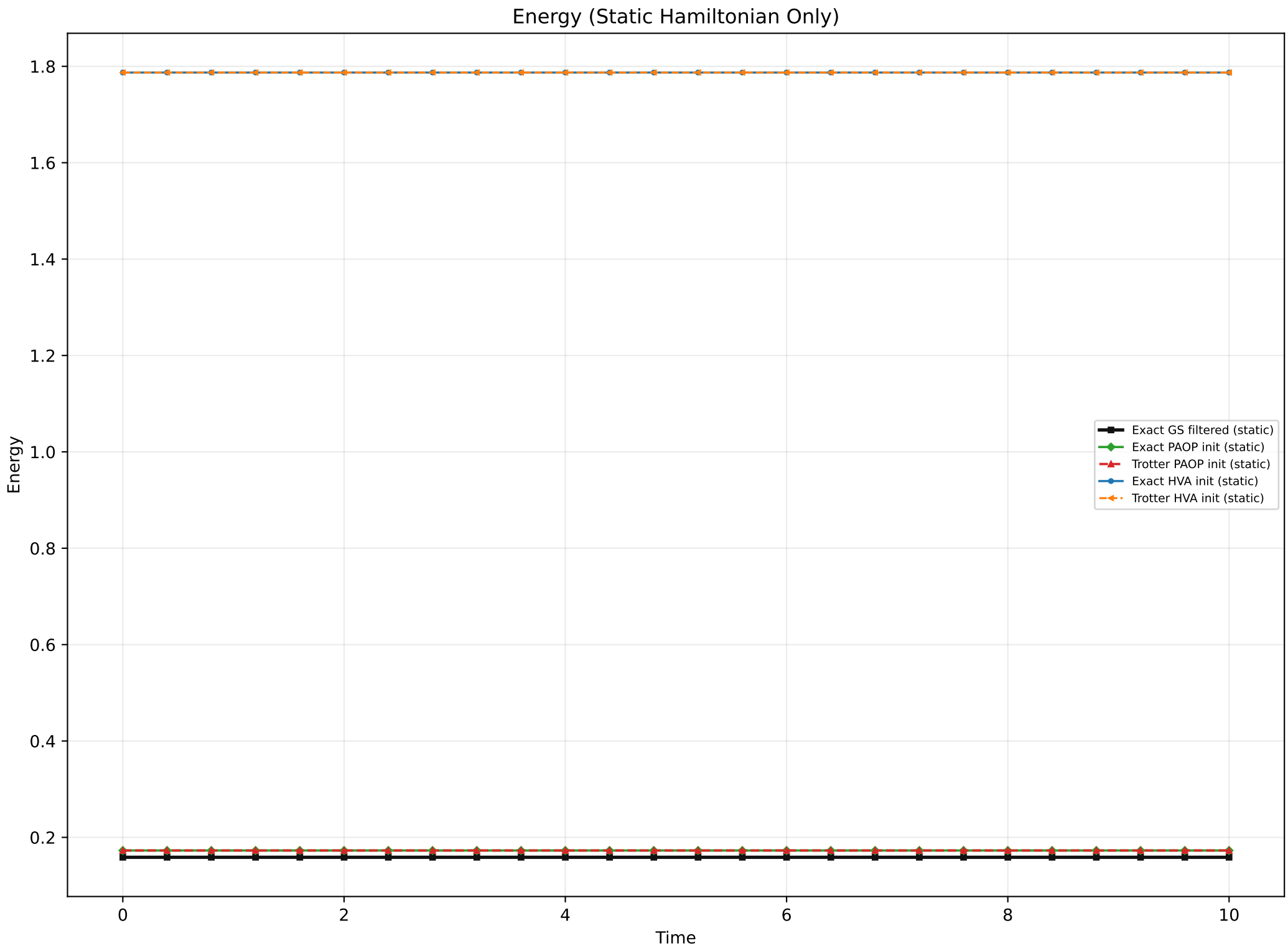


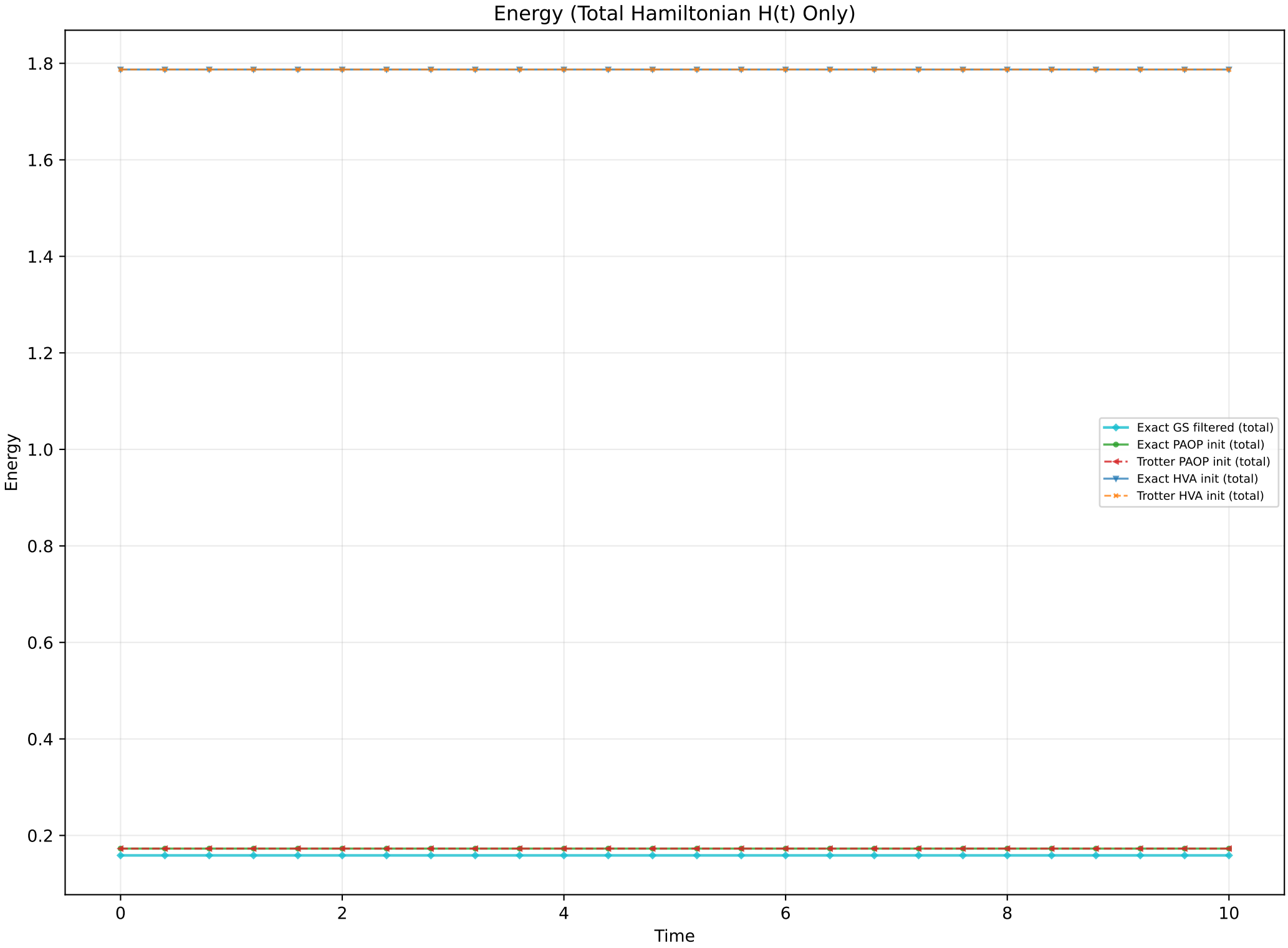

L=2 Error Heatmaps (Absolute Errors)
|n_trot(selected=hva) - n_exact(selected=hva)|
|n_exact(selected=hva) - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1e−14
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0
2
4
6
8
10
2.0
1.5
1.0
0.5
0.0
0.30
0.25
0.20
0.15
0.10
0.05

L=2 Scalar Error Heatmap




Hardcoded Hubbard Pipeline: L=2
ity(t): projected fidelity of trotterized PAOP/HVA branches vs filtered exact GS manifold.
Fidelity (PAOP trotter)
Fidelity (HVA trotter)
Energy
Exact GS filtered (static)
Exact PAOP init (static)
Trotter PAOP init (static)
Exact HVA init (static)
Trotter HVA init (static)
Site-0 Occupations
n_up0 exact GS
n_up0 exact selected (hva)
n_up0 trotter
n_dn0 exact GS
n_dn0 exact selected (hva)
n_dn0 trotter
Time
Total Doublon
doublon exact GS
doublon exact PAOP
doublon trotter PAOP
doublon exact HVA
doublon trotter HVA
Time

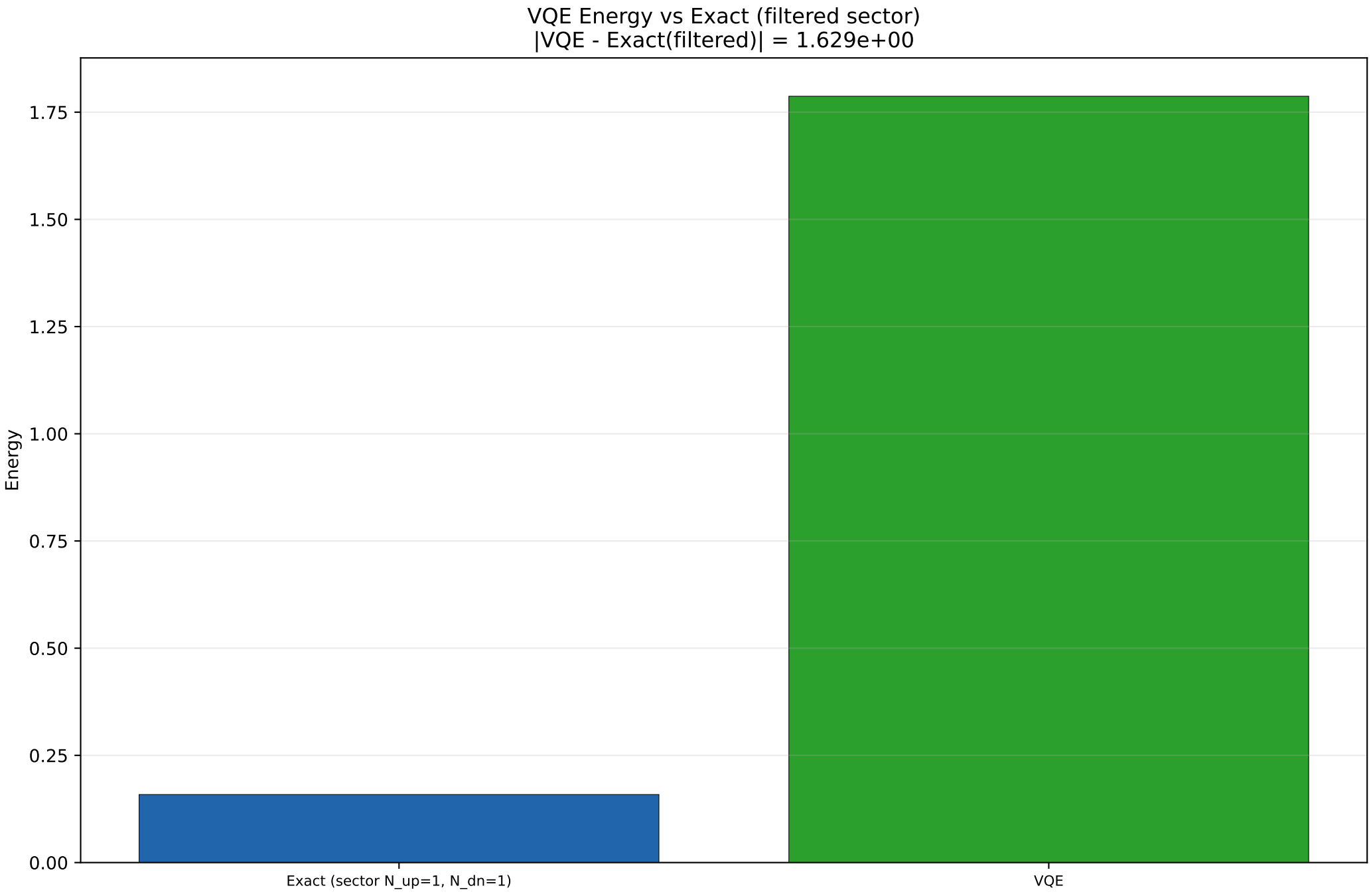

Energy (Static Hamiltonian Only)
Energy
Time
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
2
4
6
8
10
Exact GS filtered (static)
Exact PAOP init (static)
Trotter PAOP init (static)
Exact HVA init (static)
Trotter HVA init (static)


Energy (Total Hamiltonian H(t) Only)
Energy
Time
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
2
4
6
8
10
Exact GS filtered (total)
Exact PAOP init (total)
Trotter PAOP init (total)
Exact HVA init (total)
Trotter HVA init (total)

Appendix — L=2

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

Ansatz:
  - hardcoded ansatz: hh_hva
  - vqe method: hardcoded_hh_hva_layerwise_statevector

Energy + Fidelity:
  - subspace_fidelity_at_t0: 0.3625670619623603
  - subspace_fidelity_paop_t0: 0.991372243326367
  - subspace_fidelity_hva_t0: 0.3625670619623603
  - energy_t0_exact_gs: 0.15866790412572584
  - energy_t0_exact_paop: 0.17276768699322725
  - energy_t0_trotter_paop: 0.17276768699322734
  - energy_t0_exact_hva: 1.7871733127680274
  - energy_t0_trotter_hva: 1.7871733127680276
  - ground_state_exact_energy_full_hilbert: -0.150680123886
  - ground_state_exact_energy_filtered: 0.15866790412572704
  - filtered_sector: {'n_up': 1, 'n_dn': 1}
  - vqe_energy: 1.7871733127680272
  - qpe_energy_estimate: None

Config:
  - initial_state_source: vqe
  - exact_trajectory_label: Exact_Hardcode
  - exact_trajectory_method: python_matrix_eigendecomposition
  - fidelity_definition: fidelity_paop_trotter(t) = <psi_paop_trot(t)|P_exact_gs_subspace(t)|psi_paop_trot(t)> and
    fidelity_hva_trotter(t) = <psi_hva_trot(t)|P_exact_gs_subspace(t)|psi_hva_trot(t)>, where
    P_exact_gs_subspace(t) projects onto the time-evolved filtered-sector ground manifold selected by E <= E0 +
    tol. Legacy key fidelity(t) follows the selected initial-state branch.
  - hamiltonian_terms: 17
  - reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```